Justin Bunker

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Professional Summary

Machine learning researcher completing PhD at the University of Cambridge with peer-reviewed publications, including in the Journal of Machine Learning Research, and international conference presentations. Expertise in generative models and deep learning applied to synthetic data generation, computer vision, and scientific computing. Previous experience developing proprietary trading software at DRW and quantitative analysis tools at Société Générale. Strong foundation in both machine learning theory and practical software engineering in financial environments.

Education

University of Cambridge — **Cambridge, UK**
PhD in Machine Learning — *2020–2026*

- Thesis: *Generative Models in Function Space and Applications to Synthetic Data Augmentation* (in preparation)
- Supervised by Prof. Mark Girolami
- **Splunk Scholar** – Fully funded by Splunk Inc.
- Published "Autoencoders in Function Space" in the Journal of Machine Learning Research (JMLR)
- Applied diffusion models to synthesize training data for improved segmentation in remote sensing with limited labels
- Research areas: Generative models (VAEs, normalizing flows, diffusion), deep learning, neural operators
- Collaborated with domain experts in remote sensing and presented research at international conferences
- Treasurer, Canadian Society at Cambridge (2023–2024)

University of Cambridge — **Cambridge, UK**
MPhil in Machine Learning and Machine Intelligence — *2018–2019*

- Thesis: *Extending and Applying the Gaussian Process Autoregressive Regression Model*
- Supervised by Prof. Richard E. Turner
- Coursework in probabilistic machine learning, deep learning, computer vision, and optimization
- Group project: Deep learning methods for continual learning

Concordia University — **Montreal, QC**
Independent Student (Part-time) — *2017–2018*

- Coursework in stochastic methods, mathematical statistics, and statistical simulation
- GPA: 4.30/4.30

Concordia University — **Montreal, QC**
Bachelor of Computer Engineering, Great Distinction — *2012–2016*

- **Computer Engineering Medal** recipient
- GPA: 4.04/4.30
- Russell Breen Scholarship recipient
- Developed an AI algorithm that won an in-class competition

Dawson College — **Montreal, QC**
DEC in Electronics Engineering Technology — *2009–2012*

- Specialization in Computers and Networks
- Graduated with honors

Professional Experience

University of Cambridge — **Cambridge, UK**
Research Assistant — *2019–2020*

- Collaborated with Splunk Inc. on machine learning research for clustering machine-generated log data
- Developed a variational Bayes mixture model with coreset summarization and user feedback (labels and pairwise constraints)
- Published results in *Pattern Recognition Letters*

Vigilant Global – A DRW Company **Montreal, QC**
Software Developer *2016–2018*
- Developed features for proprietary trading software in an event-driven trading environment
- Designed and implemented full-stack training system for candidate evaluation in simulated trading scenarios
- Refactored legacy codebases to improve performance and integrate modern frameworks
- Applied quantitative and analytical skills to build tools supporting systematic trading strategies

Société Générale **Montreal, QC**
Software Developer (Co-op) *2015*
- Implemented features in C# for collateral monitoring application in trading operations
- Migrated financial reports to SSRS framework, managing full development lifecycle
- Presented technical solutions to business stakeholders in the Corporate and Investment Banking division

Ericsson **Montreal, QC**
Software Developer (Co-op) *2013*
- Developed Python automation scripts for OSS network benchmarking
- Reduced data extraction time from 8 hours to 2 hours through process optimization (75% improvement)

Publications

2025: **Bunker, J.**, Girolami, M., Lambley, H., Stuart, A.M., & Sullivan, T.J. "Autoencoders in Function Space." *Journal of Machine Learning Research*, 26, 1–54.

2024: **Bunker, J.**, Hadjidemetriou, G.M., d'Avigneau, A.M., & Girolami, M. "On the performance of pothole detection algorithms enhanced via data augmentation." *Transportation Research Procedia*, 78, 230–237.

2022: **Bunker, J.**, Curtis, K., Girolami, M., & Sriharsha, R. "A mixture modeling approach for clustering log files with coreset and user feedback." *Pattern Recognition Letters*, 156, 74–80.

Technical Skills

Programming: Python, C/C++, SQL

ML Frameworks: PyTorch, JAX, scikit-learn

Specialized Skills: Generative models (VAEs, normalizing flows, diffusion), neural operators (FNOs, DeepONets), statistical learning, deep learning, optimization, data analysis

Tools & Systems: Git, Linux, LaTeX, Jupyter

Awards & Scholarships

2020–2025: **Splunk Scholar** – Full PhD funding from Splunk Inc.

2016: **Computer Engineering Medal** – Top graduating student, Concordia University

2013: **Russell Breen Scholarship** – Academic excellence, Concordia University

Selected Presentations

2025: *"Synthetic SAR Imagery Generation Using Diffusion Models for Ice Floe Monitoring"* – EGU25 Conference, Vienna, Austria

2024: *"Autoencoders in Function Space"* – Invited Talk, Physics X, London, UK

2023: *"On the Performance of Pothole Detection Algorithms Enhanced via Data Augmentation"* – EWGT2023, University of Cantabria, Santander, Spain

2023: *"Introduction to JAX for Machine Learning"* – Internal Symposium, University of Cambridge, Cambridge, UK

2022–2024: *"Introduction to Machine Learning for Civil Engineering Researchers"* – Annual seminar, University of Cambridge, Cambridge, UK

Teaching Experience

University of Cambridge **Cambridge, UK**
Supervisor *2022–2024*

- Supervised teaching sessions for **3M1: Mathematical Methods**, Part IIA Engineering Tripos (annual)
- Topics: Linear algebra, stochastic processes, optimization methods

MEng Project Supervision..........

2024: **Shuohan Tao** – “Enhancing the Normalizing Flow on the Function Space”

2024: **Prithvi Raj** – “Learning Gradient Variance Control with Thermodynamic Integration”

2021: **Jay Wong** – “Design of Experiments for Bayesian Partition Models”

Languages

English: Native speaker

French: Native speaker *Fully bilingual*